The structural characteristics of pre-industrial economy

(Alfani-Colli, ch. 1-2.1)

30067 - Economic history

A.y. 2025/2026

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In this class

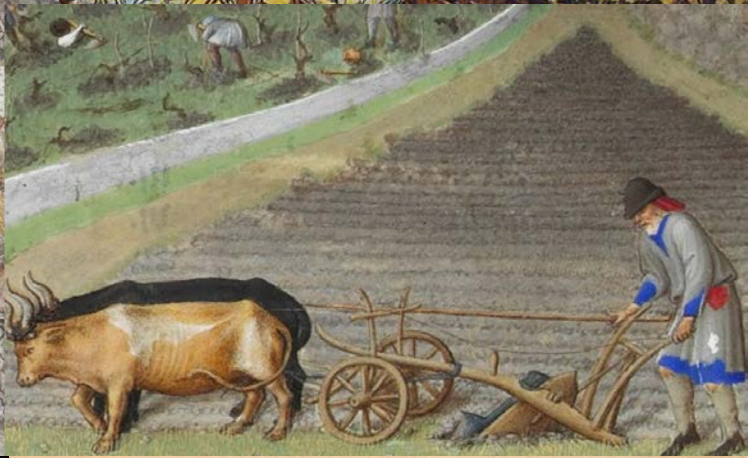
- The shift from hunting and gathering to agriculture
    - The agricultural revolution (or Neolithic revolution)
    - The urban revolution
- The structural features of agrarian societies
- The Malthusian theory of population growth
- An accelerator of history: the Black Death

These societies

- Can you interpret the economic aspects characterizing them?
- Can you highlight differences?
  - Institutions
  - Demography
  - Consumption and Savings
  - Technological progress
  - Resources allocation

- How and why did this happen?
- What factors in place?



Would you say they are different?

First of all, agriculture

Before the Industrial Revolution, societies were agricultural societies.

- Agriculture and animal husbandry were the backbone of the entire socio-economic structure.

Understanding the industrialization process requires grasping the significance of the **discovery of agriculture**

- the foundation for settled societies
- surplus food production
- the development of technology and specialization

All of which were essential for the transition to industrialized economies

Moreover, agriculture has often changed the environment and landscape in a decisive way.

First of all, agriculture

Attention! We must not take agriculture for granted.

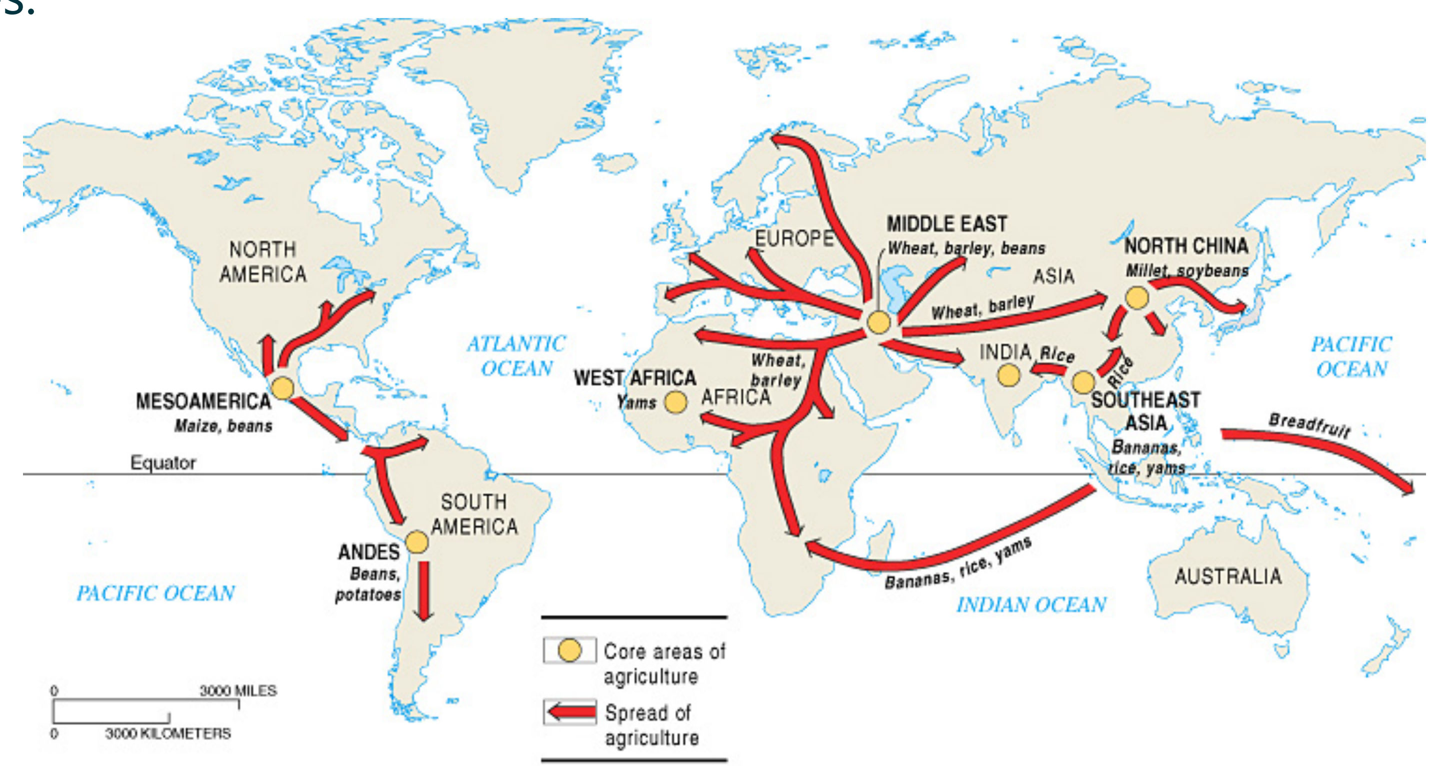
Agriculture was in fact an invention, one of the most important in human history. Without agriculture, mankind would not have been on the right track to achieve (centuries and centuries later) industrialisation.

The agricultural revolution

The invention of **agriculture was not a phenomenon confined to a single region of the planet**. During roughly the same period, humans in at least seven distinct areas of the globe engaged in agricultural practices.

In different regions, various species were domesticated based on:

availability, climate, etc.

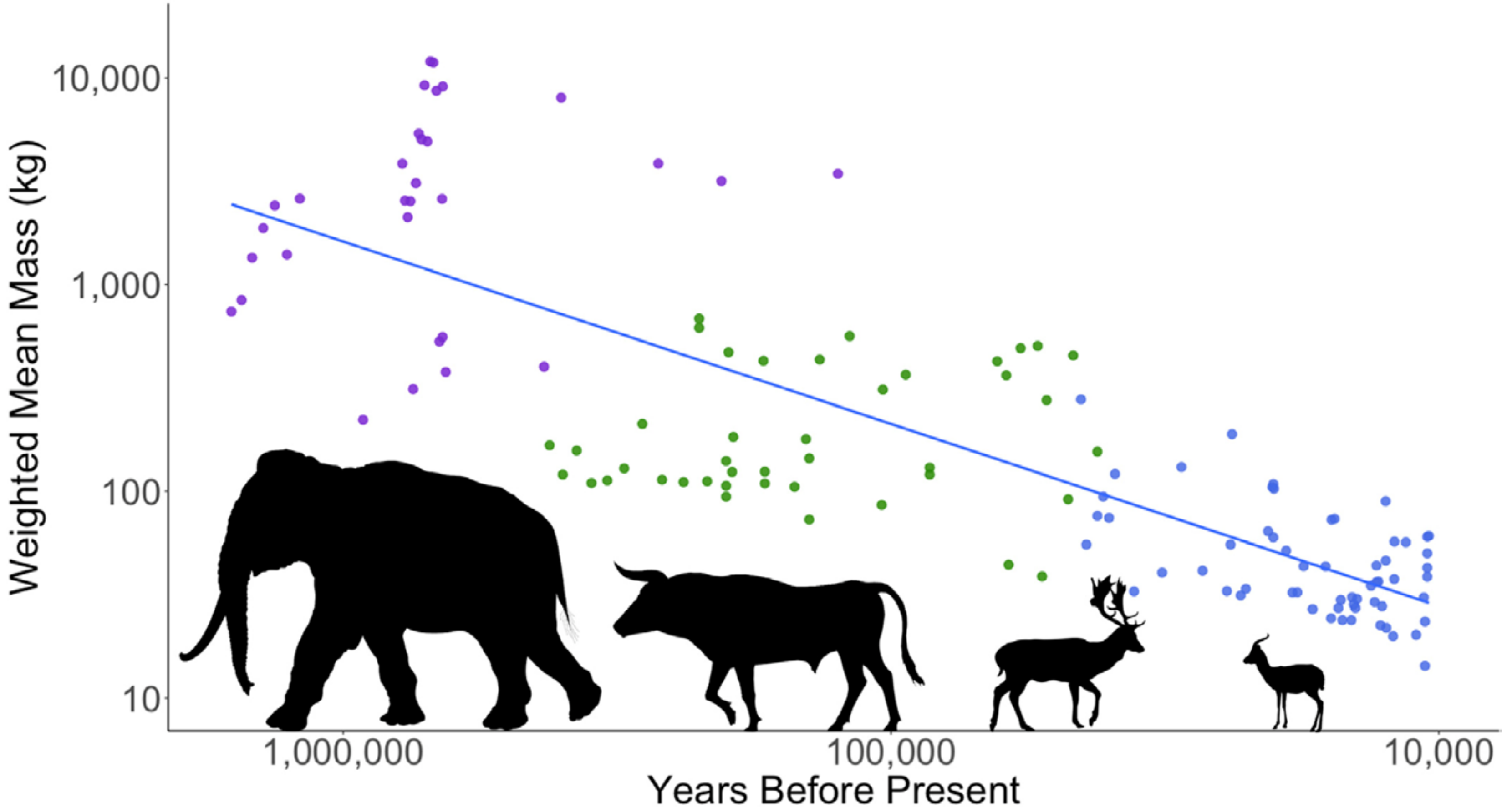


Why there was an agricultural revolution in the Neolithic?

- According to the classical theory, humans “**discovered**” agriculture and became farmers just adopting their crucial invention.
- Some recent interpretations argue instead that men and women were **forced** into change by demographic pressure.

The assumption is that a number of techniques had been, slowly, **assimilated** simply by **observing** nature and that they had been placed in the body of knowledge of a particular population in the expectation that they would be convenient to use.

What is the reletionship b/w resources and humans?



Dembitzer, Jacob, et al. "Levantine overkill: 1.5 million years of hunting down the body size distribution." *Quaternary Science Reviews* 276 (2022): 107316.

The shift to agriculture: a mixed blessing

Agricultural Revolution (10.000 BCE)

Pros:

- More food available
- Increase in population
- Increase in socio-economic complexity

Cons:

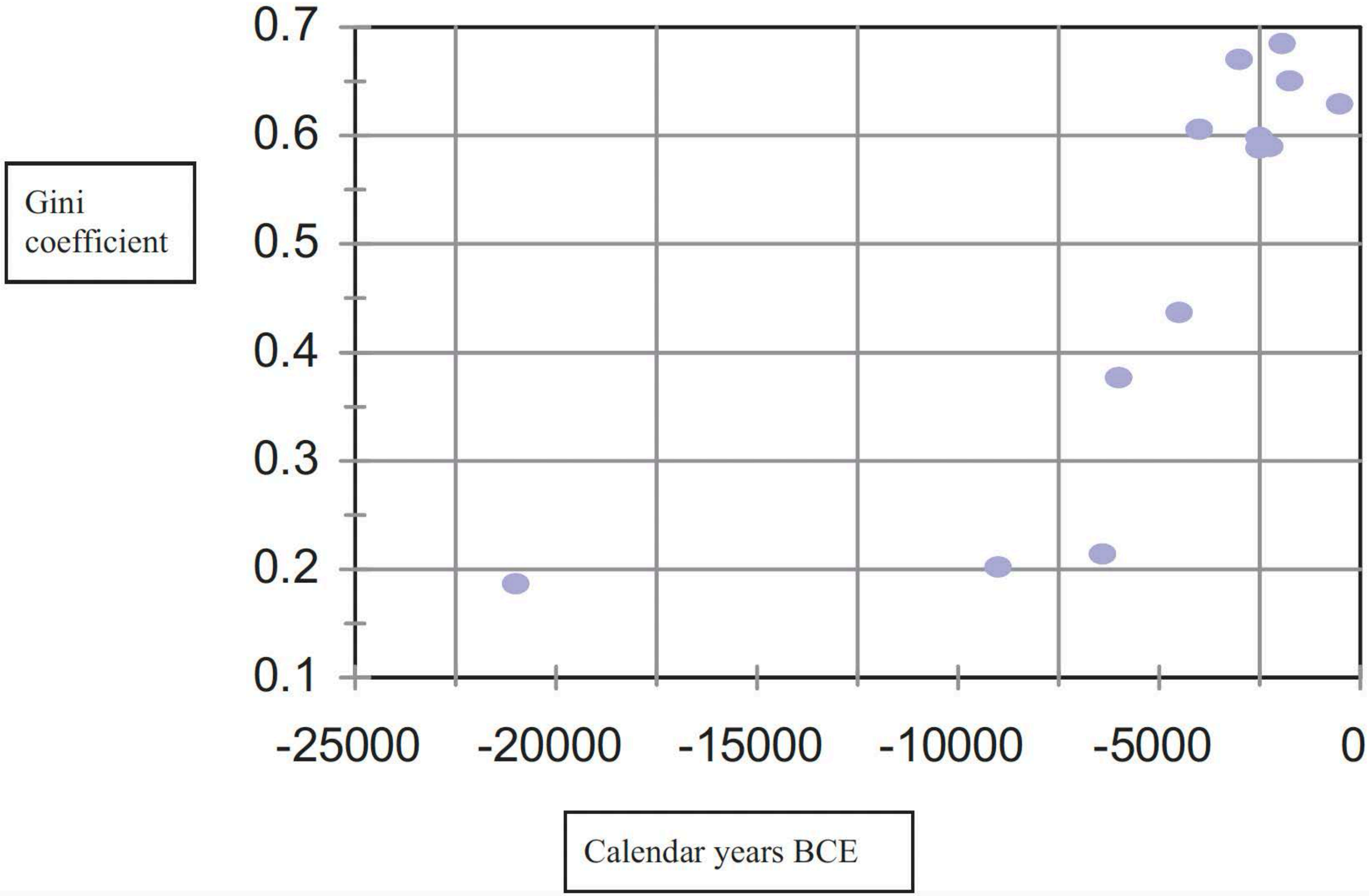
- More pressure on natural resources
- Less variety in diet → worst nutritional intake
- More diseases (also due to the proximity of people in villages)

The shift to agriculture: a mixed blessing

Table 1.1 Population, total births, and years lived (10,000 BCE to 2000 CE).

Demographic index	10,000 BCE	0	1750	1950	2000
Population (millions)	6	252	771	2,529	6,115
Annual growth (%)	0.008	0.037	0.064	0.594	1.766
Doubling time (years)	8,369	1,854	1,083	116	40
Births (billions)	9.29	33.6	22.64	10.42	5.97
Births (%)	11.4	41.0	27.6	12.7	7.3
Life expectancy (e_0)	20	22	27	35	56
Years lived (billions)	185.8	739.2	611.3	364.7	334.3
Years lived (%)	8.3	33.1	27.3	16.3	18.0

Notes: For births, life expectancy, and years lived, the data refer to interval between the date at the head of the column and that of the preceding column (for the first column the interval runs from the hypothetical origin of the human species to 10,000 BCE).



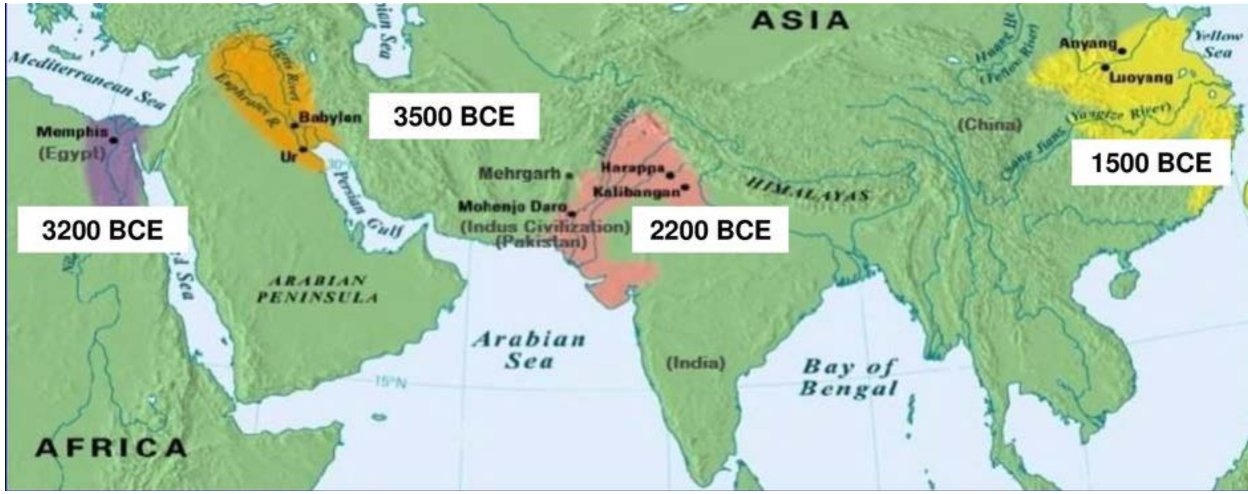
Robert C. Allen (2024).The Neolithic Revolution in the Middle East: A survey and speculation article for the Economic History Review, 1-43

From agriculture to…

However, it was **during the Bronze Age that we witnessed the acceleration of urbanization and the development of complex societies**.

The mastery of metallurgy, particularly bronze, facilitated advances in tools, weaponry, and infrastructure, promoting specialized labour and enabling the construction of fortified cities. These urban centres became hubs of trade, governance, and cultural innovation, giving rise to the earliest city-states and empires in human history.

The positive impact of agriculture paved the way for another major historical development: the **urban revolution** of the Bronze Age. Compared to the agricultural revolution, the urban revolution was a purely Eurasian phenomenon.



The urban revolution (3000 BCE)

The Neolithic Agricultural Revolution laid the foundation for:

→ **surplus food production**

→ stable communities.

population growth

emergence of **social hierarchies**.

Urban Revolution (3000 BCE)

Pros:

- Higher division of labour
- Accumulation of knowledge
- Development of complex socio-political structures typical of “modern state” forms

Cons:

- Higher taxation
- Increase in inequality

Urban revolution and proto-divergence

- The (agri-)urban revolution was a typical Eurasian phenomenon and thanks to its gains fostered a kind of proto-divergence between Eurasia itself and the rest of the world.
    - A more productive agriculture
    - Higher population density
    - More complex state institutions

Why Eurasia?

Instutional explanation (Jack Goody)

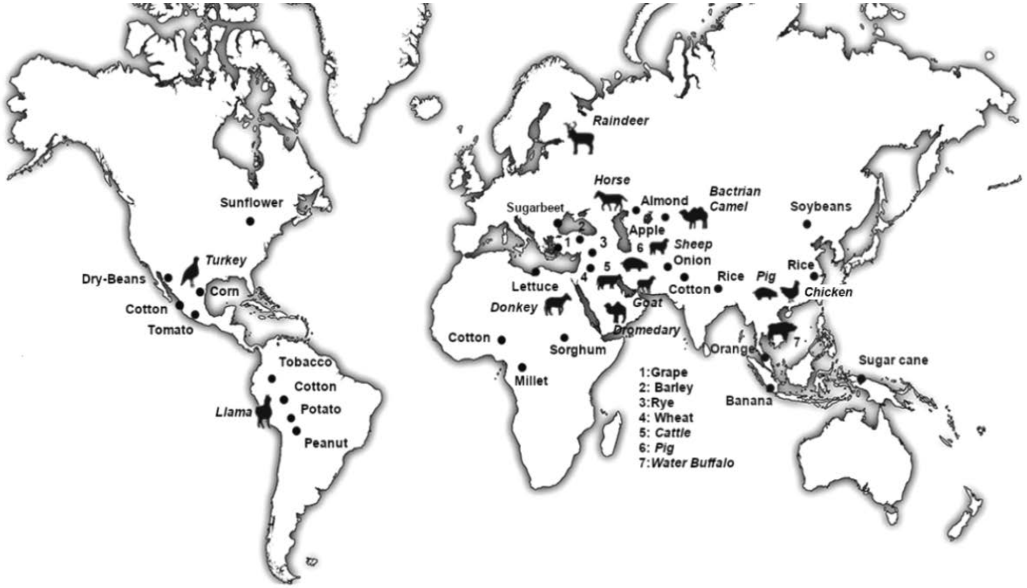
- Stronger political, economic, and social institutions → State capacity
- Recognition of private property
- Stricter inheritance systems
- Stronger role of the family

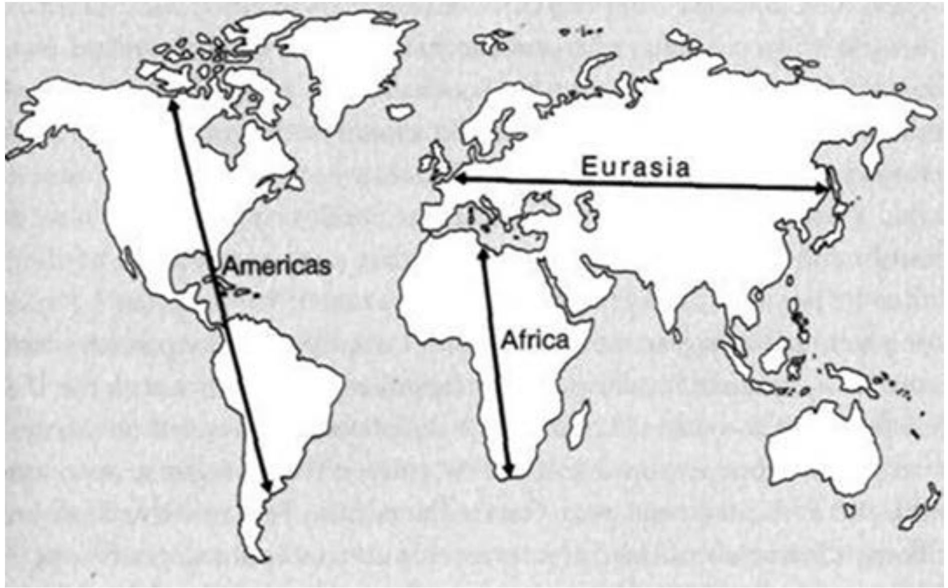
Urban revolution and proto-divergence

Environmental explanation (Jared Diamond)

- In Eurasia, there would be:

- **More and better species** both animal and vegetable
- **Geographical orientation** E-W more favourable to human and crop migration
- **Fewer natural barriers**
- A pre-existing advantage was linked to the **earlier spread of the human species** in Eurasia from Africa.





Features of agrarian societies

Although agrarian societies were (greatly) more complex than hunter-gatherer societies, their complexity was (greatly) less than that of post-industrial societies. Why?

- **Low urbanization**
  - 6-8% in Western Europe, 20-25% in Italy before the Black Death
- **Limited division of labour**
  - based more on age and gender than on ability
- **High rate of self consumption**
  - near to 90%, with the other 9% sold in the city and 1% exported
- **Low productivity** → limited surplus
- **Strong solidarity system**
  - especially through matrimonial strategies

Features of agrarian societies: technology

Little complexity does not, however, mean a complete stasis of agrarian societies.

→Particularly from the year 1000 CE onwards (but with some significant contributions visible as early as the 7th and 9th centuries).

European societies underwent considerable progress, especially from a technological point of view:

- Heavy plow
- Collar harness for animals
- Three-year crop rotation
- Water-mills

Human Capital

Physical Capital

Whether "hardware" or "software", technology enhances labour productivity, thereby driving output growth

The role of the city….
and the market

- In pre-modern agrarian societies, the city specialised in:
    - the production of **manufactured goods**
    - the **exchange** of goods
    - the **production and provision of services** (civil and ecclesiastical magistracies).

WHAT IS THE ROLE OF THE MARKET?

- It promotes specialization and increase productivity (Adam Smith 1776)
- In the functioning of pre-industrial economies, the **complementarity** of town and country was fundamental.
- Similarly, in pre-industrial development processes, **an advancement of all economic sectors** (agriculture, manufacturing, trade) was essential.

The most important cities (e.g. Venice):

- emerged as poles of commercial and financial capitalism
- became pivotal centres for long-distance trade
- became important knowledge dissemination centres (technological innovations, institutional, consumer behaviour)

What is a problem for economic growth to occur in this context?

Resilience and vulnerability of agrarian economies

- Among the most significant characteristics of agrarian societies is certainly the precarious **relationship between people and resources**.
- One of the most famous attempts to analyse (and model) this relationship is that of Thomas Malthus, who in 1789 published his book *Essay on the Principle of Population*.

  → The idea of Malthus is that societies are not able to prosper as much as they would like because the constraints imposed by nature are essentially insurmountable.

<u>Caveat: Malthusian theory is not perfect and a pre-industrial society was not necessarily subject to it. But it remains an important tool for understanding how pre-industrial societies functioned</u>.

Resilience and vulnerability of agrarian economies

"We will suppose the means of subsistence in any country just equal to the easy support of its inhabitants. The constant effort towards population increases the number of people before the means of subsistence are increased. The food therefore which before supported seven millions, must now be divided among seven millions and a half or eight millions. The poor consequently must live much worse, and many of them be reduced to severe distress. The number of labourers also being above the proportion of the work in the market, the price of labour must tend toward a decrease; while the price of provisions would at the same time tend to rise. The labourer therefore must work harder to earn the same as he did before."

Thomas Malthus, 1798.

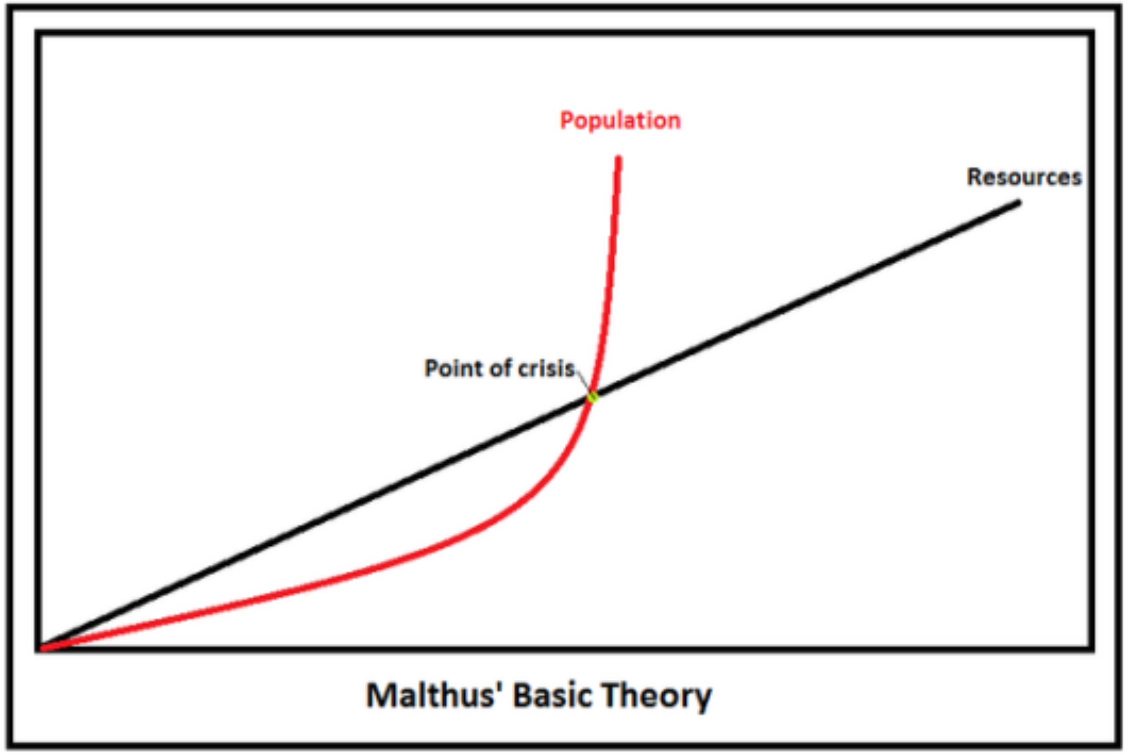
Resilience and vulnerability of agrarian economies

The basis of this idea is the principle that the population is growing in a geometrical rate: 2, 4, 8, 16, 32 etc. The food supply, on the other hand, increased only in an arithmetical fashion: 2, 4, 6, 8 etc.

Since the population will grow more quickly than the food supply. This will result in a food shortage.

"…the power of population is indefinitely greater than the power in the earth to produce subsistence for man."

Thomas Malthus, 1798.



Resilience and vulnerability of agrarian economies

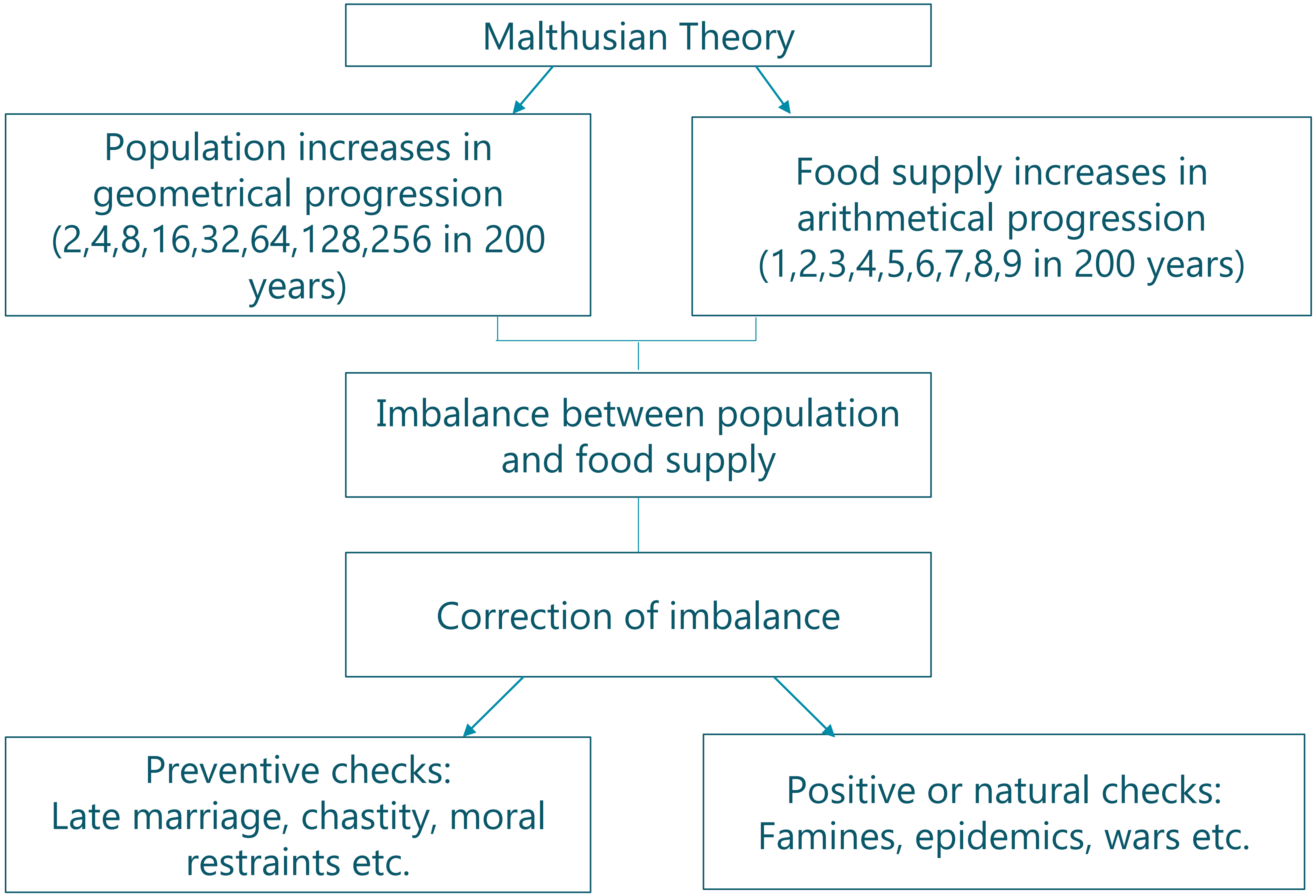
Two types of check can occur and limit population expansion:

Positive checks

- Population increase so much that the per **capita income decreases** and thus mortality increases
- Larger population will be exposed to **dramatic events** that raise the mortality rate (e.g. famines and plagues)

Preventive checks

- **Households' behaviour** aimed at reducing fertility: in pre-industrial economies delayed marriage rather than contraceptive practices
- Households will keep the **birth rate** at the subsistence level

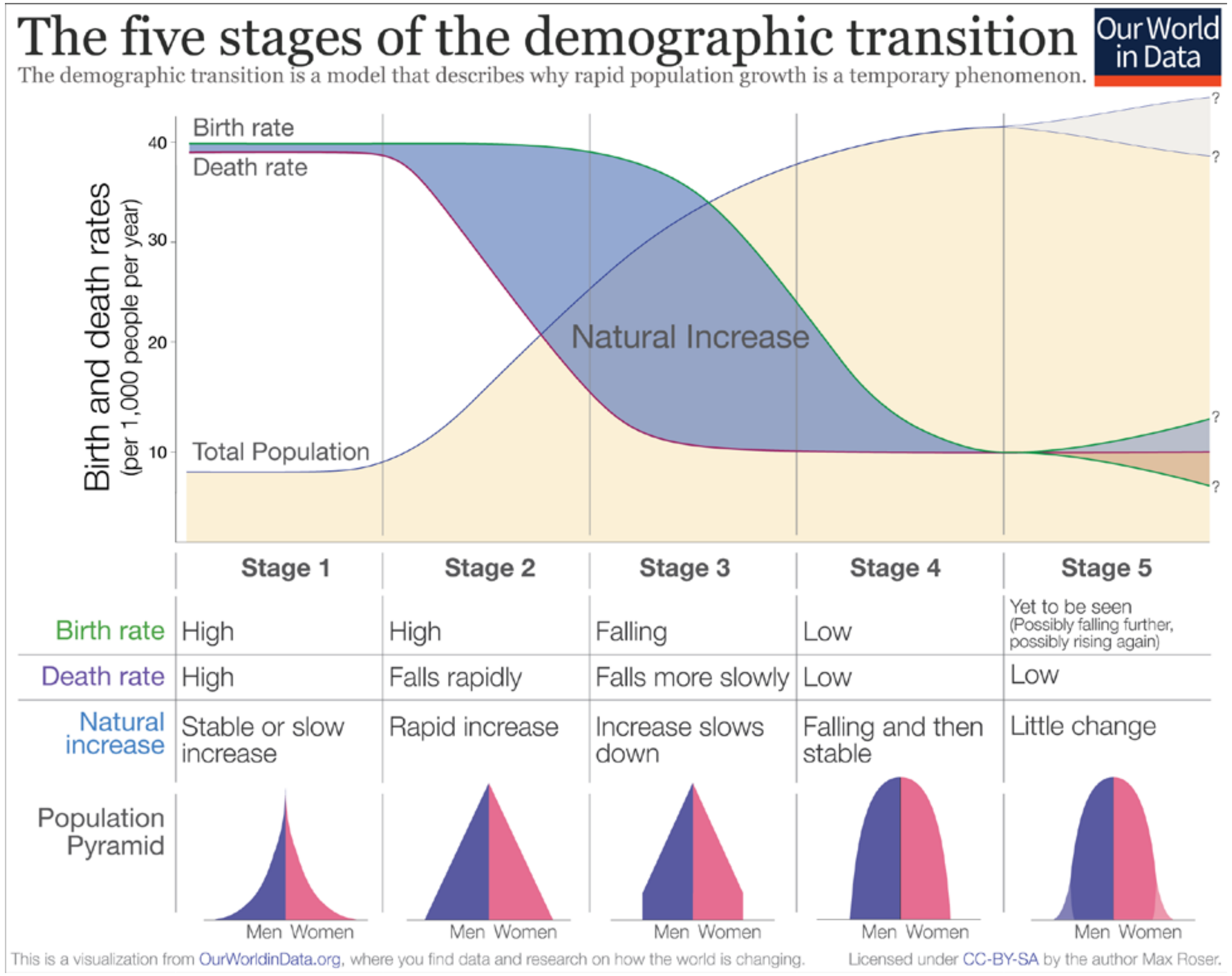

Malthusian Theory
Population increases in geometrical progression (2,4,8,16,32,64,128,256 in 200 years)
Food supply increases in arithmetical progression (1,2,3,4,5,6,7,8,9 in 200 years)
Imbalance between population and food supply
Correction of imbalance
Preventive checks: Late marriage, chastity, moral restraints etc.
Positive or natural checks: Famines, epidemics, wars etc.

Escaping the Malthusian trap

It was not until the 18th century, starting in England, that the Malthusian trap was overcome, thanks to:

- the modernisation of agriculture
- the industrial revolution

These two factors were crucial in promoting a **demographic transition**.

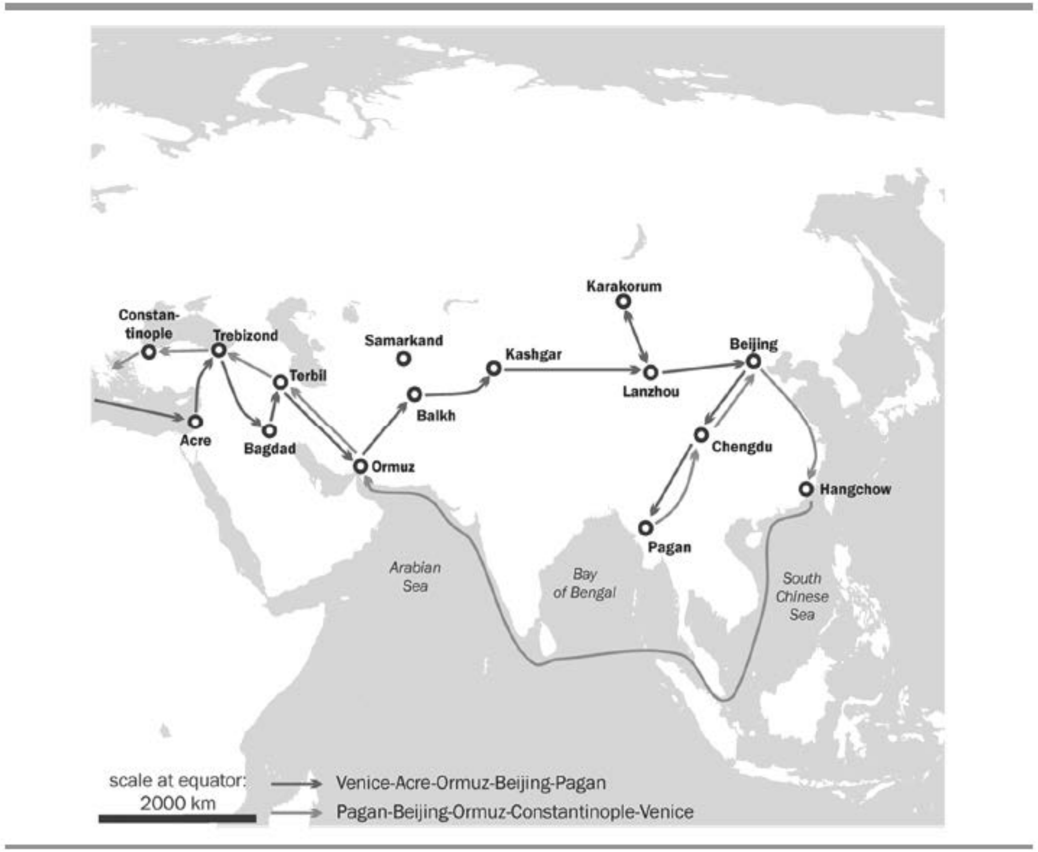


What could be an «exogenous solution»?

An accelerator of history: the Black Death

- It is true that changes in pre-industrial societies were slow (although, as we have seen, we cannot speak of a complete immobility). However, some exogenous shocks greatly accelerated development.
- The most important, in European history, was the Black Death.

Figure 2.1. – *Marco Polo's journey along the Silk Road*

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Source: *Philip's Atlas of World History.*

The Black Death: a mixed blessing

Pros:

- Less pressure on natural resources
- More land available
- Mitigation of economic inequalities (and higher social mobility)
- Improvement of living standards

Cons:

- Drastic demographic decline (between 30 and 60%) and dramatic loss of human capital
- Sharp slowdown in the production system
- Stop of international (and generally supra-regional) trade

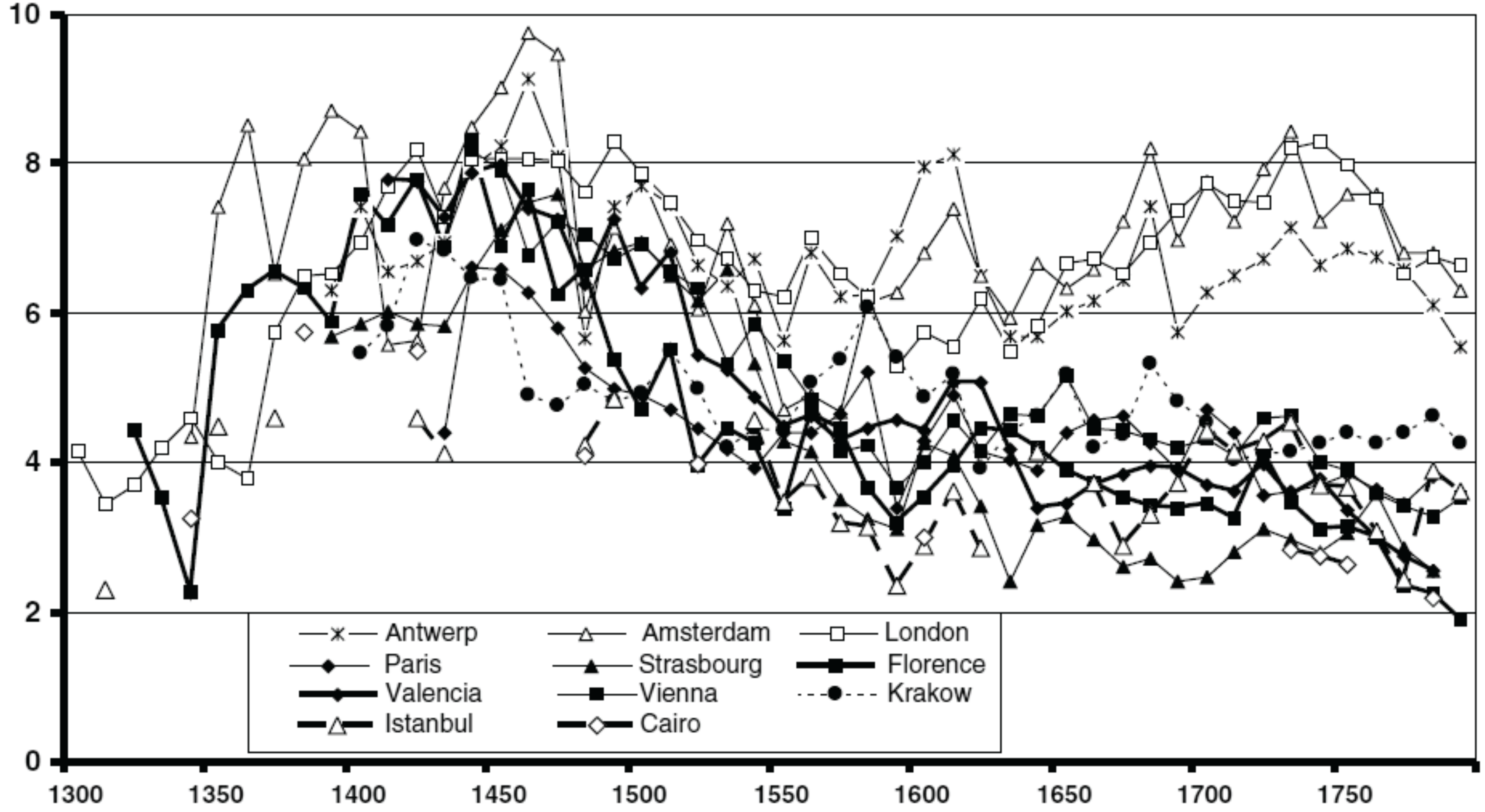
What are the economic consequences?

The effects of the Black Death

Table 1. *Population of selected European countries, 1300–1800 (in thousands)*

	1300	1400	1500	1600	1700	1800
England and Wales	5,750	3,000	3,500	4,450	5,450	9,250
Netherlands	800	600	950	1,500	1,950	2,100
Belgium	1,250	1,000	1,400	1,600	2,000	2,900
Italy	12,500	8,000	9,000	13,300	13,500	18,100
Spain	5,500	4,500	5,000	6,800	7,400	11,000
Total Europe	94,200	67,950	82,950	107,350	114,950	192,230

Source: Paolo Malanima (unpublished manuscript).



Source: S. Pamuk, The Black Death and the Origins of the 'Great Divergence' across Europe, 1300.- 1600, in European Review of Economic History, 2007, 11(3), p. 297. The graph shows real wages of unskilled workers converted to indices.

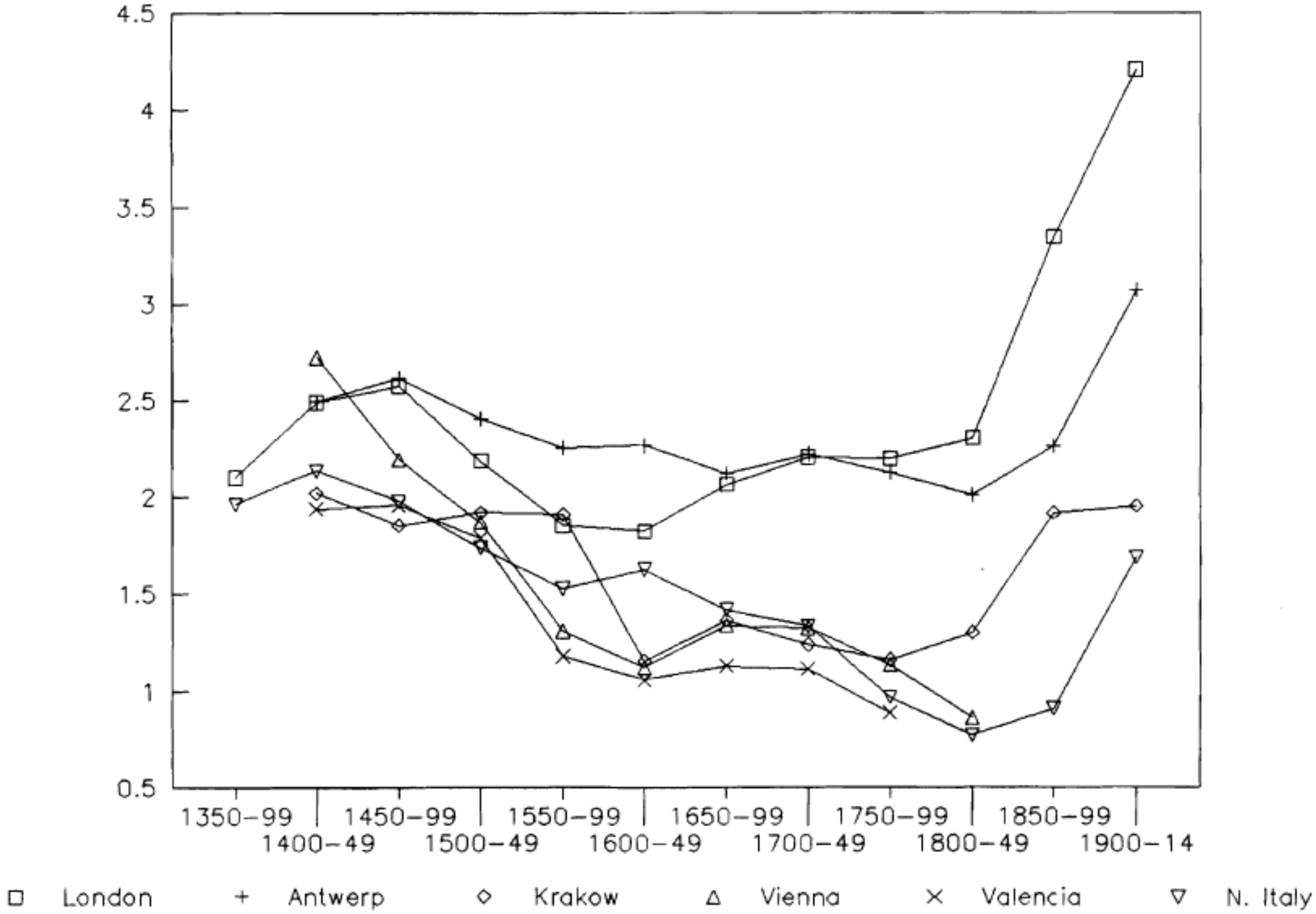


FIG. 7. Welfare ratios for craftsmen (earnings relative to poverty line).